



Practice Assignment 7 - Not Graded



This assignment will not be graded and is only for practice.

Note: Suppose W_1 and W_2 are subspaces of a vector space V . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of V .

Multiple Select Questions (MSQ):

1 point

Let $V_1 = \mathbb{R}^3$ be a vector space, define a function $\|\cdot\|_i : V_1 \rightarrow \mathbb{R}$ for $i = 0, 1, 2, 3$ as:

1. $\|(x, y, z)\|_0 = x + y + z$
2. $\|(x, y, z)\|_1 = |x| + |y| + |z|$
3. $\|(x, y, z)\|_2 = \sqrt{x^2 + y^2 + z^2}$
4. $\|(x, y, z)\|_3 = \max\{|x|, |y|, |z|\}$

and let $V_2 = M_{2 \times 2}(\mathbb{R})$ be a vector space, define a function $\|\cdot\|_i : V_2 \rightarrow \mathbb{R}$ for $i = 4, 5$ as:

1. $\|A\|_4 = \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\}$, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.
2. $\|A\|_5 = \max\{|a_{11}| + |a_{21}|, |a_{12}| + |a_{22}|\}$, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.

Which of the following option(s) is (are) true.

☐

$\|\cdot\|_0$ is a norm on V_1 .

☐

$\|\cdot\|_1$ is not a norm on V_1 .

☐

$\|\cdot\|_2$ is a norm on V_1 .

☐

$\|\cdot\|_3$ is not a norm on V_1 .

☐

$\|\cdot\|_4$ is not a norm on V_2 .

☐

$\|\cdot\|_5$ is a norm on V_2 .



No, the answer is incorrect.

Score: 0

Accepted Answers:

$\| \cdot \|_2$ is a norm on V_1

$\| \cdot \|_5$ is a norm on V_2

1 point

Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^3$, i.e.,

$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$. Consider a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as:

$$T(u) = \langle u, v \rangle, T(v) = \langle u, v \rangle,$$

where $v \in \mathbb{R}^3$. Which of the following options is (are) true for T ?

☐

T is one-one for all $v \neq 0 \in \mathbb{R}^3$.

☐

T is onto for all $v \neq 0 \in \mathbb{R}^3$.

☐

T is not one-one for every $v \in \mathbb{R}^3$.

☐

There exists a $v \in \mathbb{R}^3$ such that T is an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is onto for all $v \neq 0 \in \mathbb{R}^3$.

T is not one-one for every $v \in \mathbb{R}^3$.

1 point

Which of the following options are true?

☐

$V = \mathbb{R}^2$ and a function defined as:

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R} \\ \langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^2 x_i y_j. \langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R} = \sum_{i,j=1}^2 x_i y_j.$$

is not an inner product on the vector space V .

☐

$V = M_{2 \times 2}(\mathbb{R})$ and $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on the vector space V , a function defined as: $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ is an inner product on the vector space V .

☐

$V = M_{3 \times 3}(\mathbb{R})$ and $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on the vector space V , a function defined as: $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 - \langle \cdot, \cdot \rangle_2$ is an inner product on the vector space V .

☐

$V = \mathbb{R}^2$ and a function defined as:

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R} \\ \langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 - x_2 y_1. \langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R} = x_1 y_2 - x_2 y_1.$$

is an inner product on the vector space V .

is an inner product on the vector space V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \mathbb{R}^2$ and a function defined as:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R} \\ \langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^2 x_i y_j. \langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R} = \sum_{i,j=1}^2 x_i y_j.$$

is not an inner product on the vector space V .

$V = M_{2 \times 2}(\mathbb{R})$ and $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on the vector space V , a function defined as: $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ is an inner product on the vector space V .

1 point

Consider a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as

$T(x, y, z) = (x + 2y, x - y + cz, 2x + y + dz)$. Let A be the matrix representation of T with respect to the standard

ordered bases of $\mathbb{R}^3$ and $B = \begin{bmatrix} 0 & 3 & 2 \\ -1 & -3 & -3 \\ 2 & 3 & 4 \end{bmatrix}$.

Based on the above information which of the followings option(s) is(are) true?

☐

If A is similar to B , then rank of the linear transformation T is 2.

☐

If A is similar to B , then nullity of the linear transformation T is 2.

☐

If $c \neq d$, then A cannot be similar to B .

☐

If A is similar to B and $c = 1$, then a basis of the kernel is $\left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A is similar to B , then rank of the linear transformation T is 2.

If $c \neq d$, then A cannot be similar to B .

If A is similar to B and $c = 1$, then a basis of the kernel is $\left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}$.

1 point

Consider a vector $v = (1, 2)$ from the inner product space $\mathbb{R}^2$, where inner product is the dot product. Which of the following options is/are true?

☐

There exists a vector $u (\neq v)$ in $\mathbb{R}^2$ such that $\|u\| = \|v\|$.

☐

There are infinitely many vectors in $\mathbb{R}^2$ such that norms of the vectors are equal to norm of v .



There exists a vector u in R^2 such that $\langle u, v \rangle = 0$.



There are infinitely many vectors in R^2 such that angle between the vector and v is equal to 90° .

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a vector u ($u \neq v$) in R^2 such that $\|u\| = \|v\|$.

There are infinitely many vectors in R^2 such that norms of the vectors are equal to norm of v .

There exists a vector u in R^2 such that $\langle u, v \rangle = 0$.

There are infinitely many vectors in R^2 such that angle between the vector and v is equal to 90° .

1 point

Let L and L' be affine subspaces of R^3 , where $L = U$ and $L' = (2, 0, 1) + U'$, for some vector subspaces U and U' of R^3 . Let a basis for U be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ and a basis for U' be given by $\{(1, 0, 1), (0, 1, 1)\}$. Suppose there is a linear transformation $T: U \rightarrow U'$ such that $(0, 1, 0) \in \ker(T)$, $T(2, 0, 1) = (0, 1, 1)$ and $T(1, 1, 0) = (1, 0, 1)$. An affine mapping $f: L \rightarrow L'$ is obtained by defining $f(u) = (2, 0, 1) + T(u)$, for all $u \in U$. Which of the following options are true?



$L = \{(x, y, z) \mid x - y - 2z = 0\}$



$L = R^3$



$L' = \{(x, y, z) \mid x + y - z = 1\}$



$L' = R^3$



$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$



$f(x, y, z) = (x - 2z + 2, \frac{x}{2}, x - z + 1)$



$f(x, y, z) = (x - 2z, z, x - z)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L = R^3$

$L' = \{(x, y, z) \mid x + y - z = 1\}$

$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$

1 point

Let $L_1 = \{(x, y, z) \mid x + y + 2z = 6\}$ and

$L_2 = \{(x, y, z) \mid 2x + y + 2z = 9\}$ be affine subspaces of R^3

and W_1 and W_2 be the corresponding vector subspaces of the given affine subspaces of L_1 and L_2 respectively. Choose the set of correct options



$W_1 = \{(x, y, z) \mid x + y + z = 0\}$

☐

$$W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\} \quad W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$$

☐

$$W_1 = \{(x, y, z) \mid x + y + 2z = 0\} \quad W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$$

☐

$$W_2 = \{(x, y, z) \mid 2x + y + 2z = 1\} \quad W_2 = \{(x, y, z) \mid 2x + y + 2z = 1\}$$

☐

$$W_1 = \{(x, y, z) \mid x + y + 2z = 1\} \quad W_1 = \{(x, y, z) \mid x + y + 2z = 1\}$$

☐

$$W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\} \quad W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\} \quad W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$$

$$W_1 = \{(x, y, z) \mid x + y + 2z = 0\} \quad W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$$

$$W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\} \quad W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$$

Numerical Answer Type (NAT):

Let θ be the angle between the vectors $u = (2, 1, -3)$ and $v = (5, 8, 4)$ and

$\|\cdot\|$ denote the usual length in R^3 with respect to the dot product, then what is

$$\|u\| \|v\| \cos(\theta) \quad \|u\| \|v\| \cos(\theta)?$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on R^3 , i.e.,

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3 \quad \langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$$

Let $v_1 = (1, 0, -1)$, $v_2 = (a, b, c) \in R^3$. Suppose the length of the vector

$$v_2 \text{ is } 3 \text{ and } \langle v_2, (1, -1, 1) \rangle = -3, \langle v_2, (-1, 2, 1) \rangle = 1$$

If the angle between v_1 and v_2 is 45° , then find the value of $a + b + c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Let $A_{4 \times 4}$, $B_{4 \times 4}$ and $C_{4 \times 4}$ be matrices such that AA and BB are equivalent matrices and BB and CC are similar matrices. If rank of the matrix AA is 3 then find the determinant of the matrix CC .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point